Topic 10: Clock Tree Synthesis

ECE 9433: Agile System-on-Chip Design

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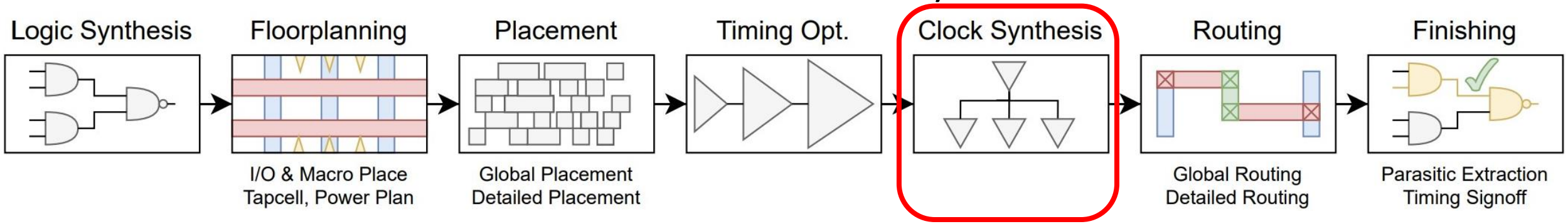
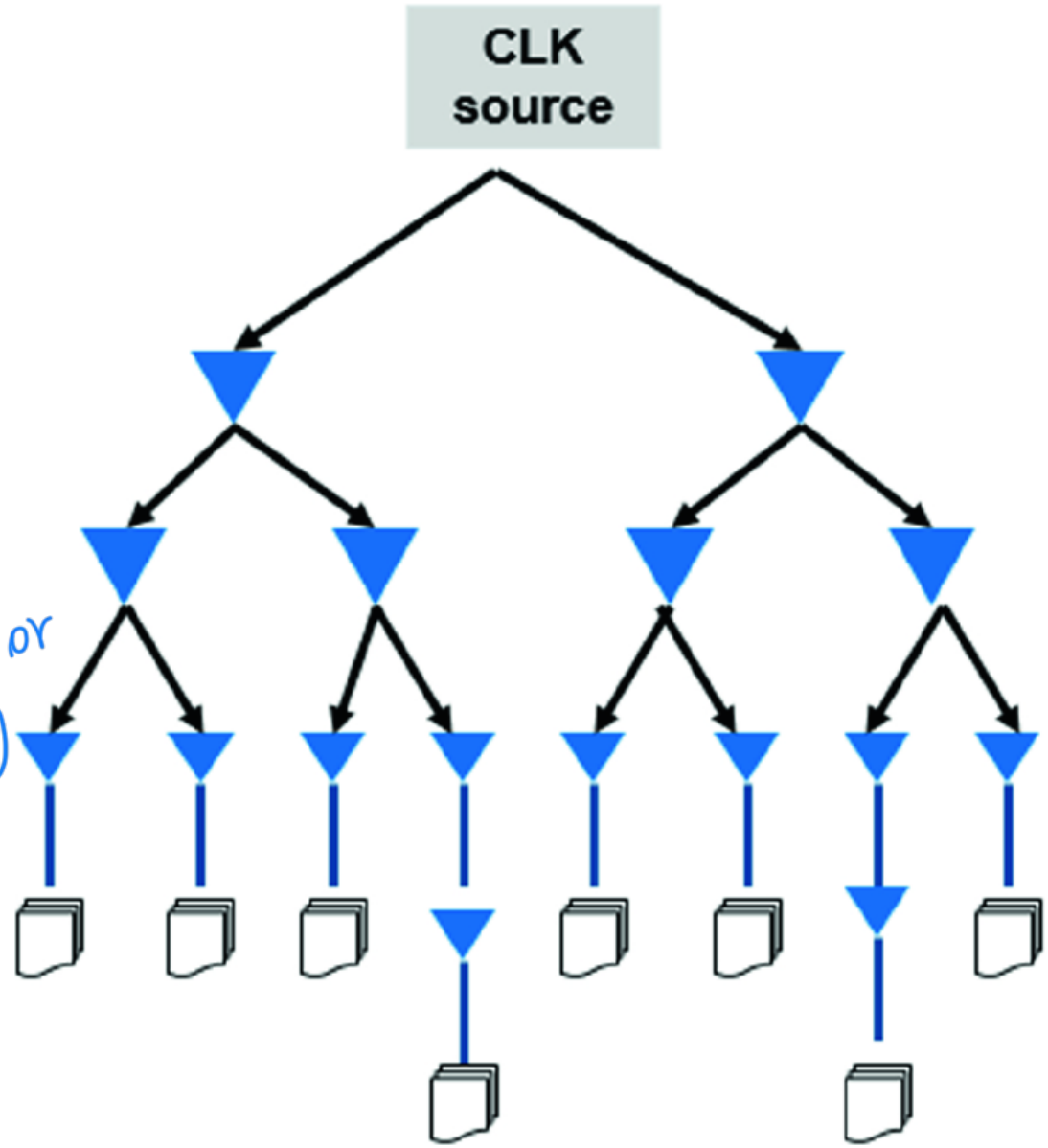
Distribute clock as close as possible



Clock Tree Synthesis

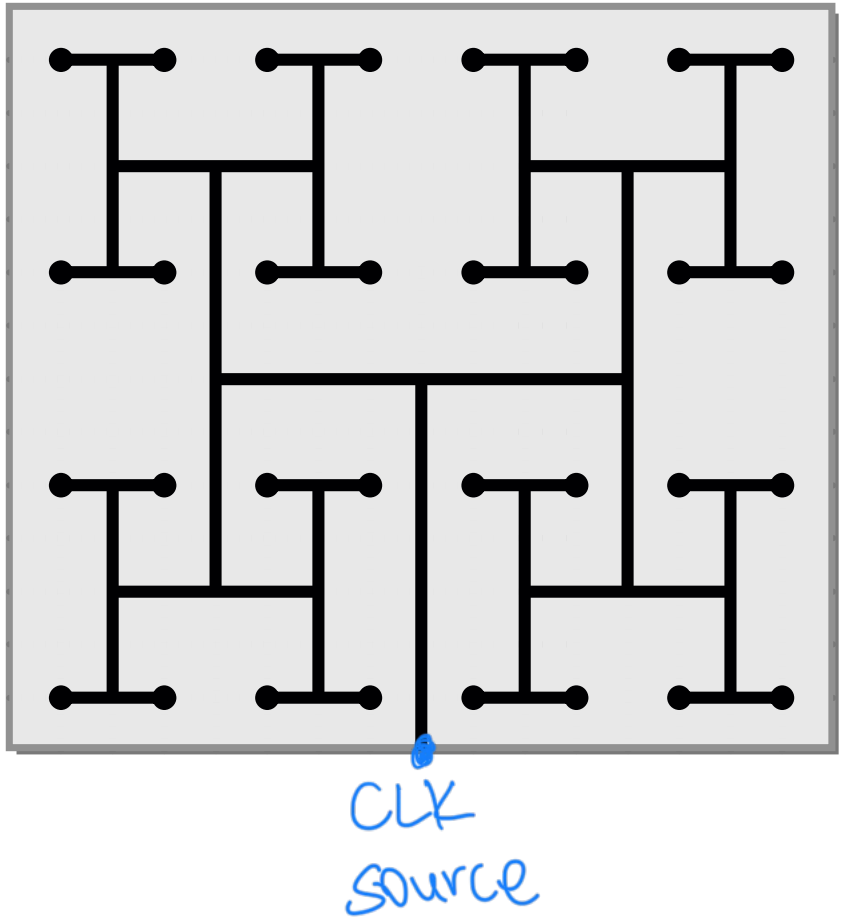
- Clock signals must arrive roughly at the same time at all *sinks / endpoints*
- **Clock sink**: terminal or I/O where the clock signal terminates
  - Most often a register, but could also be a macro (e.g. SRAM) or output
- Clock tree synthesis optimizes ~~between~~ **skew** and **power** more buffer or more wiring
  - Skew only needs to be good enough to pass timing
  - **Useful skew**: purposefully skewing endpoints to improve timing

balanced skew



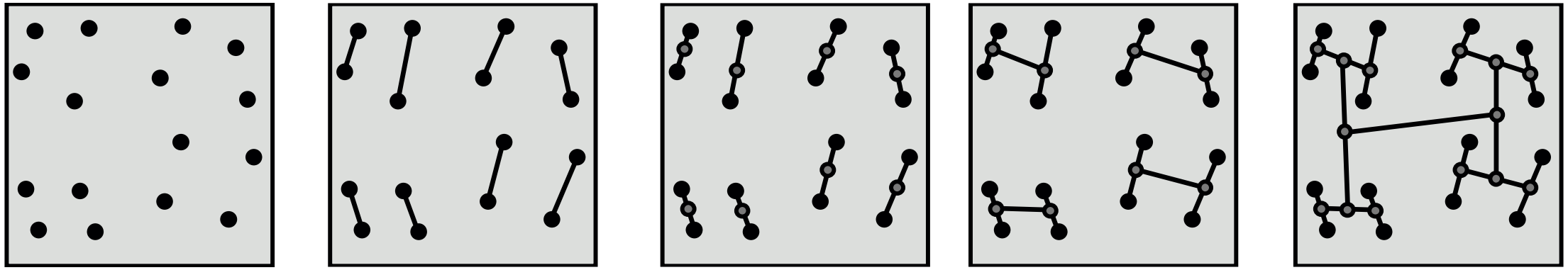
Clock Distribution Challenge

- Time from source (input pin) to sink (register, etc.) is dependent on wirelength and buffering
- In advanced technologies, variation can play a heavy hand
- Ideal, zero-skew network is an H-tree
  - Exact same wirelength and number of buffers to reach every sink
  - Very inefficient! Lots of wirelength spent to ensure matching in every branch



Clock Tree Synthesis Algorithms

- Other methods can achieve a better balance of skew and power
  - E.g. Recursive Geometric Matching (RGM)
- Most CTS algorithms cluster sinks into groups then build up a tree
- Bottom-up or hybrid bottom-up/top-down algorithms have proven best historically
- Try to make number of clusters, number of levels equal for whole tree



RGM algorithm

Clock Tree Synthesis: User Perspective

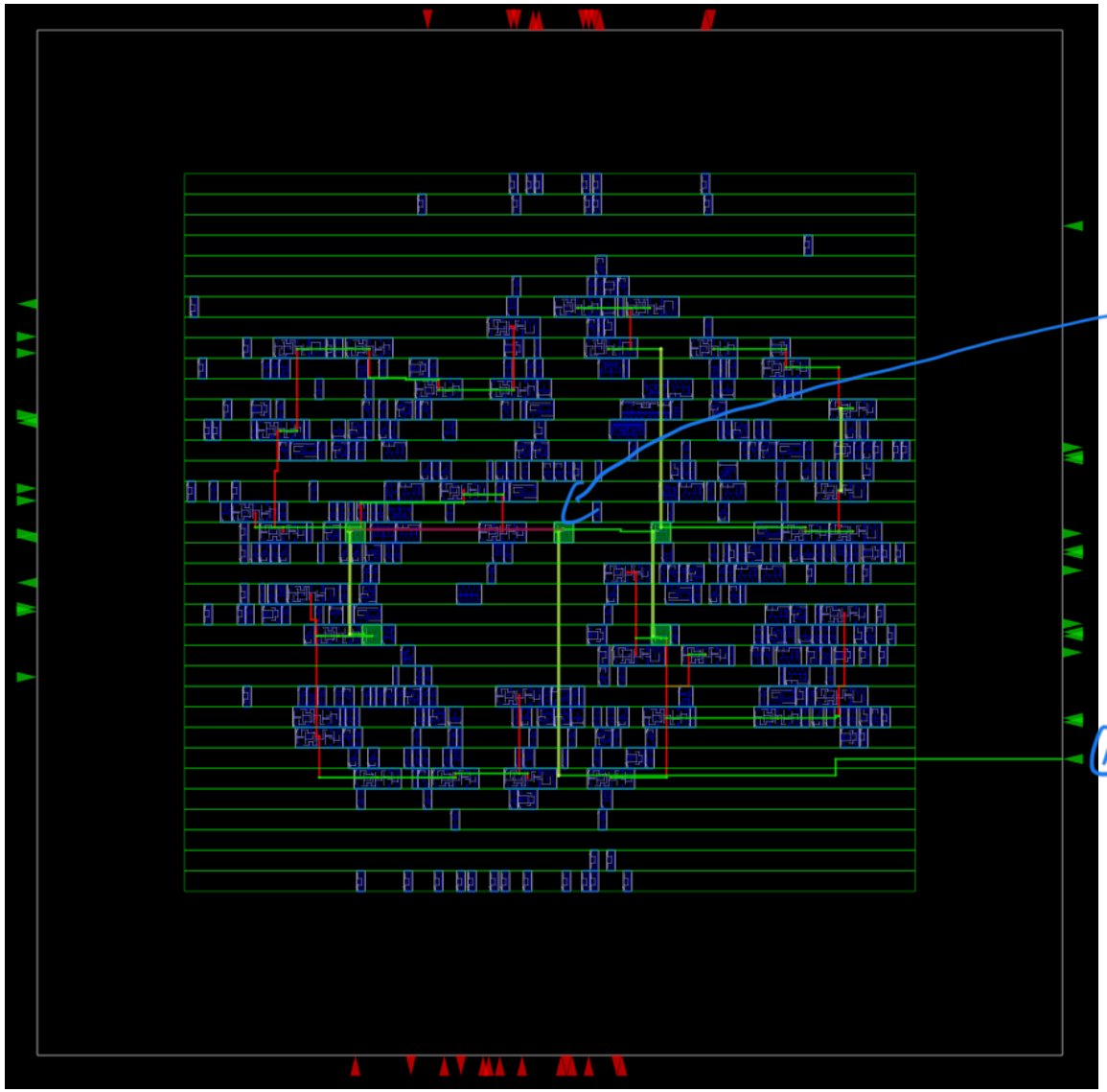
- User provides list of buffers
- Pre-characterization decides buffers, target wirelengths, etc.
- CTS performs clustering, buffer insertion, and (usually) routing
- Timing optimization follows to fix hold time after clock tree is built and skew is known

```
[INFO CTS-0049] Characterization buffer is: BUF_X4.
[INFO CTS-0039] Number of created patterns = 11880.
[INFO CTS-0084] Compiling LUT.
Min. len    Max. len    Min. cap    Max. cap    Min. slew   Max. slew
2           8           1           34          1           14
```

```
[INFO CTS-0007] Net "clk" found for clock "core_clock".
[INFO CTS-0010]  Clock net "clk" has 35 sinks.
[INFO CTS-0008] TritonCTS found 1 clock nets.
[INFO CTS-0097] Characterization used 1 buffer(s) types.
[INFO CTS-0027] Generating H-Tree topology for net clk.
[INFO CTS-0028]  Total number of sinks: 35.
```

```
Repair setup and hold violations...
[INFO RSZ-0040] Inserted 3 buffers.
[INFO RSZ-0041] Resized 29 instances.
[WARNING RSZ-0062] Unable to repair all setup violations.
[INFO RSZ-0033] No hold violations found.
```

Clock Tree Synthesis: User Perspective



CTS Parameters

good for skew.

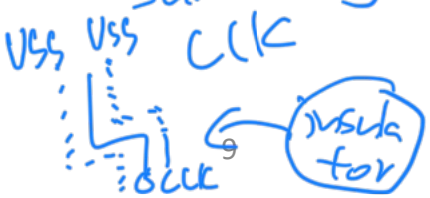
- CTS is automatic, but high-perf. designs often merit hand-tuning
- **Routing topology**: trees broken into 2-3 regions – trunk, branches, leaves. No standard distinction between the three.

Important takeaway!

- **Routing layers**: Upper metal layers better for routing global signals
  - Upper layers are thicker. $\uparrow C, \downarrow\downarrow R = \downarrow RC = \downarrow t_d$, ***but***: $\uparrow C = \uparrow CV^2 = \uparrow P$
  - Higher power, faster delay, thicker/wider = less prone to variation
  - Usually, trunk = top layers, branches = middle layers, leaves = bottom layers

delay decrease, power ↑.

- **Route width**: same as above; wider = block more routing resources
- **Route spacing**: close proximity wires can cause bad cross-talk ∝ Quadratic
- **Route shielding**: Add VSS shield around clock to absorb cross-talk (extra wire surrounding CLK



- **Sink clustering**: # sinks per cluster, diameter of cluster, etc.

How to cluster/connect groups differently.

Common CTS Problems

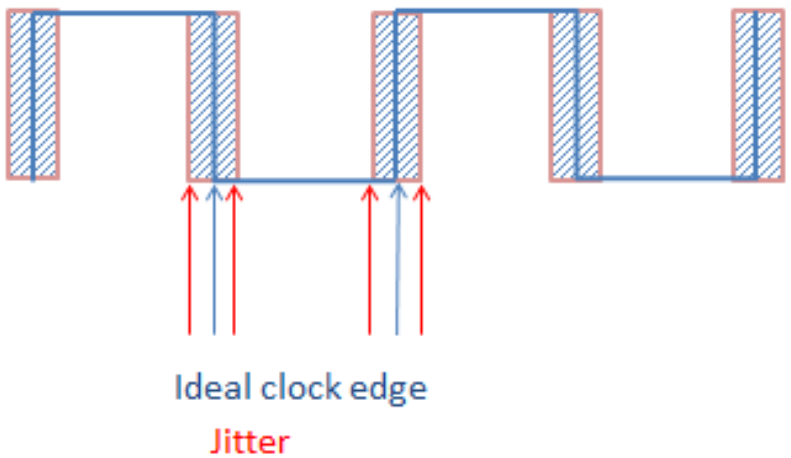
so normally multiple clock (partitioned from same clk source)

too much skew.

- Clock skew and power inherently get worse with larger trees
  - At some point, it becomes impossible to balance and other techniques must be used

there is gap between SPICE and LIBERTY file.

- Miscorrelation between estimated parasitics and extracted parasitics
  - High-performance designs use SPICE simulations to confirm CTS integrity!
- **Clock jitter**: transient fluctuations in clock
  - Causes: unstable clock source,
- Process variation
  - Variation of RC of certain routes
  - Variation of drive strength/V_t in certain buffers



Know unknown → 2% clock jitter.
Unknown unknown → no idea what % of clk jitter.

Prof. Rovinski's Favorite Question

- *If you had to fail either setup or hold time, which would you prefer?*
- Reminder

  Clock (adjustable)

  - Setup slack: $\boxed{T_c} - \boxed{t_{\mathrm{cq}} - t_{\mathrm{pd}} - t_{\mathrm{setup}} + \phi}$ ← Fixed
  - Hold slack: $\boxed{t_{\mathrm{cq}} + t_{\mathrm{cd}} - t_{\mathrm{hold}} - \phi}$ ← Fixed
- Setup time! You can simply lower your clock frequency
- Hold slack is fixed* once the chip is manufactured
  - * *Might* be able to fiddle with voltage and temperature
  - Unlikely to work, and definitely not robust